

Dynamic Model of a Rotary Kiln for Process Simulation and Control

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Abstract. Many metallurgical processes include heterogeneous reactions, where the reacting species are captured in different physical phases. Typically a loose solid phase is contacted by a gas-flow in a rotary-kiln or multiple-hearth-furnace. A mathematical model of such a thermo-chemical system was developed to describe mass-flow and temperature of gas and bulk and the concentration of the main species. For the actual application simple empirical relations were set up for the source-drain-terms, so that a model of the full process is now available. Validation showed excellent consistency with real process-data for the transport behaviour and the thermodynamic model. To validate the chemical model further work is in progress.

1. Introduction

For process-simulation and process-control it is often necessary to determine a dynamic model of the process. For the actual application a physically based model was set up to study various process modifications and also for model-based controller-design .

The actual roasting-process takes place in a multiple-hearth-furnace (see Figure), where the bulk-media is fed at the top of the furnace while the gas as a heating media and reactive partner enters from different inlets in counterflow to the bulk from bottom to top. Transportation of the bulk media is carried out by a rotating shaft with bladed arms so that the media is transported from outside to inside in one stage and back in the following stage, and so on.

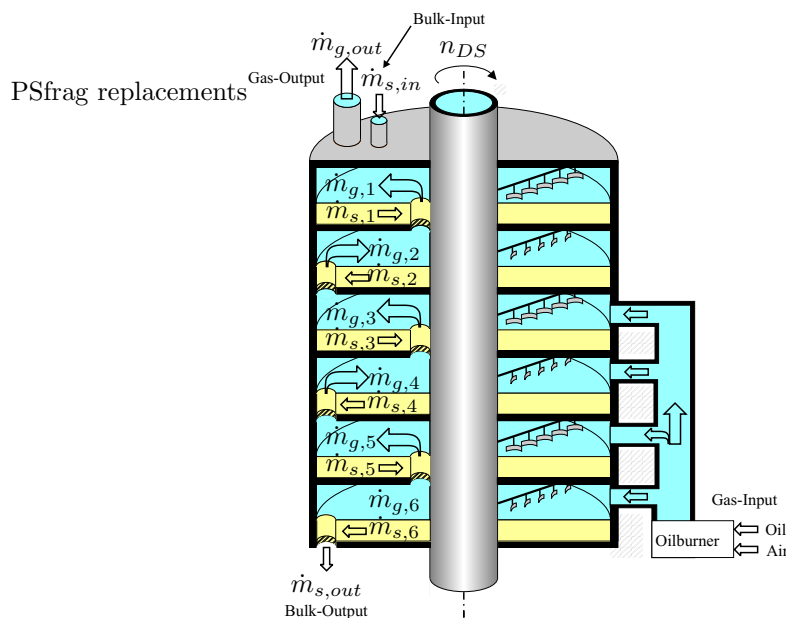


Figure 1: Scheme of a multiple-hearth-furnace

The bulk-media has to be kept within the range of the optimal process-temperature for a certain residence-time. Furthermore the concentrations of the reactive partners have to be large enough to ensure a good conversion.

2. Modelling

For mathematical modelling each floor of the furnace is divided into two ideally stirred layers, one containing the bulk-media and the other one the gas. According to the principles of a continuously stirred tank-reactor balance-equations are set up for mass, heat and the main species. Therefore the complete model of the multiple-hearth-furnace can be divided into three parts: a transportation model (mass-balances of all floors), a thermodynamic model (heat-balances of all floors) and a chemical model (mole-balances of all floors). To obtain adequate expressions for the source/drain terms two major effects are considered: drying of the bulk-media and the major chemical reaction.

Drying Kinetics:

Drying kinetics is described by a very simple two-phase drying-rate-law derived from practical drying-experiments. Therefore, the loss of weight over time of a sample taken directly from the real process was measured for 3 different temperatures. The experiments showed that a two-phase model is adequate to approximately describe the drying-process.

In drying-period I the drying-rate is assumed to be constant and independent of the drying-temperature (only valid for a very small range). This period represents the well known constant drying-rate-period (CRP), where enough surface water is available and evaporation is only depending on the environmental conditions [1][3][4]. After the water content w_{H_2O} has dropped below an empirically derived value of $w_{I,II}$ the drying-rate is assumed to be proportional to the moisture-content in drying-period II. Now the drying-rate is also depending on the mass-transport from inside the porous media to its surface. This leads to the rate-law according equations 1 and 2:

$$\frac{dw_{H_2O}}{dt} = -v_I, \quad |w_{H_2O} < w_{I,II} \quad (1)$$

$$\frac{dw_{H_2O}}{dt} = -k \cdot w_{H_2O}, \quad |w_{H_2O} > w_{I,II} \quad (2)$$

Chemical Reaction Kinetics:

As the literature survey for the very special process brought no useable result concerning the reaction-rate-law again practical experiments are the basis for the chemical reaction-kinetics rate-law. Instead of using rate-laws for every single step in the reaction-mechanism, where usually there are many of them in a heterogenous reaction-system, one global rate-law is derived, neglecting minor effects and summing up the important steps [2]. Therefore, roasting experiments were carried out under typical reactive conditions. The resulting reaction-rate-law for the major reaction is given in Equation 3.

$$r_V = k_0 e^{-\frac{E}{RT_s}} c_{ed} - k'_0 e^{-\frac{E'}{RT_s}} c_{prod}^2; \quad (3)$$

Although this is a very simple rate-law it describes the dynamic behaviour of the concentration for standard conditions well and it was also an objective of this work to see if this rate-law is all embracing enough for the stated tasks.

Transportation Model:

Transportation of the bulk-media is described by a global mass-balance for the bulk-layer. The input-term equals exactly the mass-flow exiting the previous floor. The output-term on the other hand is considered to be directly proportional to the rotation-speed of the drive-shaft. There are no source- or drain-terms considered here as a first approximation.

To take into account that at every shifting operation a part of the material remains untouched or falls back a reflux is introduced. Therefore, an effective mass-flow through the floor is defined and the outlet mass-flow can be calculated according to Equation 4. Furthermore, material densities are assumed to remain constant at floor-specific values. From that the differential-equation for the floor-mass can be reduced on the bulk-media-height in the floor according to Equation 5.

$$\dot{m}_{s,out} = (1 - f_{reflux}) \dot{m}_{eff,out} \quad (4)$$

$$\frac{h_s}{dt} = \frac{\dot{m}_{s,in}}{\rho_s A} - \frac{n_{ds}(1 - f_{reflux})h_s b}{60f_{trans}(r_o - r_i)} + \frac{\sum \dot{m}_{source} - \sum \dot{m}_{drain}}{\rho_s A} \quad (5)$$

The mass-transport of the gas-layer contains input- and output-terms and also source/drain-terms.

Thermodynamic Model:

The following effects are considered within the heat-balance for the bulk-layer: convective heat due to mass-exchange, heat of reaction, heat of evaporation, heat-transfer from the gas-layer to the bulk layer, heat-transfer to the environment, heat-transfer to the following floor through the floor plate, heat-transfer to the drive-shaft and heat for the melting of fusible components. Material-constants are assumed to remain constant. From that the heat balance according to Equation 6 can be derived, expressed as differential equation for the bulk-temperature T_s .

$$\begin{aligned} \frac{dT_s}{dt} = & u_s \frac{T_{s,in} - T_s}{r_o - r_i} + \frac{(-\Delta H_r)r_V}{c_{ps}\rho_s} + \Delta H_{ev} \frac{dw_{H_2O}}{dt} + \alpha_{GS} \frac{T_g - T_s}{c_{ps}\rho_s h_s} - \\ & - \lambda_W \frac{2r_o(T_s - T_E)}{c_{ps}\rho_s b_W(r_o^2 - r_i^2)} - \lambda_W \frac{T_s - T_{g,i+1}}{c_{ps}\rho_s h_D h_s} - \alpha_{DS} \frac{2r_i(T_s - T_{DS})}{c_{ps}\rho_s(r_o^2 - r_i^2)} - \sum_i h_{M,i} \frac{\dot{c}_i M_i}{c_{ps}\rho_s} \end{aligned} \quad (6)$$

The description of the gas-temperature is carried out in the same way, only that there are some differences concerning the heat-effects. Terms considered are convection due to mass-transfer of the gas, heat-transfer from the gas-layer to the bulk-layer, heat-transfer to the environment, heat-transfer to the previous floor through the upper floor-plate and heat-transfer to the drive-shaft. In Equation 7 the heat-balance for the gas-layer is expressed in terms of the gas-temperature T_g .

$$\frac{dT_g}{dt} = u_g \frac{T_{g,ein} - T_g}{r_a - r_i} - \alpha_{HW} \frac{2r_i(T_g - T_{HW})}{c_{pg}\rho_g(r_a^2 - r_i^2)} - \alpha_{gs} \frac{T_g - T_s}{c_{pg}\rho_g h_g} - \lambda_W \frac{2r_a(T_g - T_A)}{c_{pg}\rho_g b_W(r_a^2 - r_i^2)} - \lambda_W \frac{T_g - T_{s,i-1}}{c_{pg}\rho_g h_D h_g} \quad (7)$$

Chemical Model:

For every component taking part in the main-reaction a mole-balance is set up containing the input- and output-terms, for the source/drain term the previously described reaction-rate-law is used. In case of the mole-balance for bulk water the drying-rate-law is applied.

$$\frac{c_i}{dt} = \frac{c_{i,in}\dot{m}_{eff,in} - c_i\dot{m}_{eff,out}}{V_s\rho_s} - r_V \quad (8)$$

Modelling the multiple-hearth-furnace in the way described above resulted in a differential equation-system containing 11 equations for every floor. Since the range of the various state-variables within this system differs strongly the differential-equation-system was scaled referring to typical set-point-values. Furthermore, numerical linearization was carried out for controller-design.

3. Validation

Various experiments were carried out on the original plant to allow detailed stationary and dynamic validation of the model.

Validation of the Transportation Model:

To validate the transportation-model for the bulk-layer the bulk-mass-flow at the outlet of the multiple-hearth-furnace after a complete shutdown was recorded. Figure 2 shows the simulation-result and the measurement-result. As can be seen the simulated transport-behaviour fits not only qualitatively well but also quantitatively.

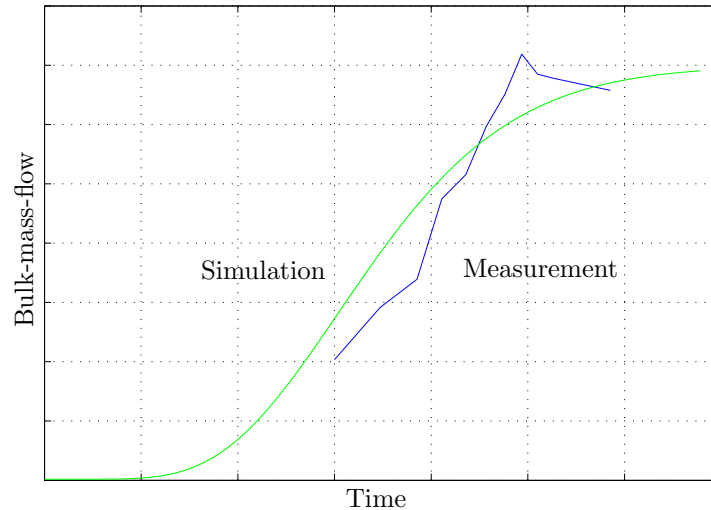


Figure 2: Mass-flow – Time for refilling of the furnace after shutdown

Validation of the Thermodynamic Model:

First of all it is necessary for the thermodynamic model to meet the stationary temperature-profile over the furnace. This means that the set point temperature of every single floor should be met approximately. Figure 3 shows the temperature-profiles of the bulk-temperature from simulation and measurement, for which the model was stationary optimized. The agreement therefore is very good, what is essential for the bulk-temperature because of the strong temperature-influence on the reaction within the bulk-media. Figure 4 shows the gas-temperature-profiles over the furnace from simulation and measurement, where the agreement is now not so good but still acceptable.

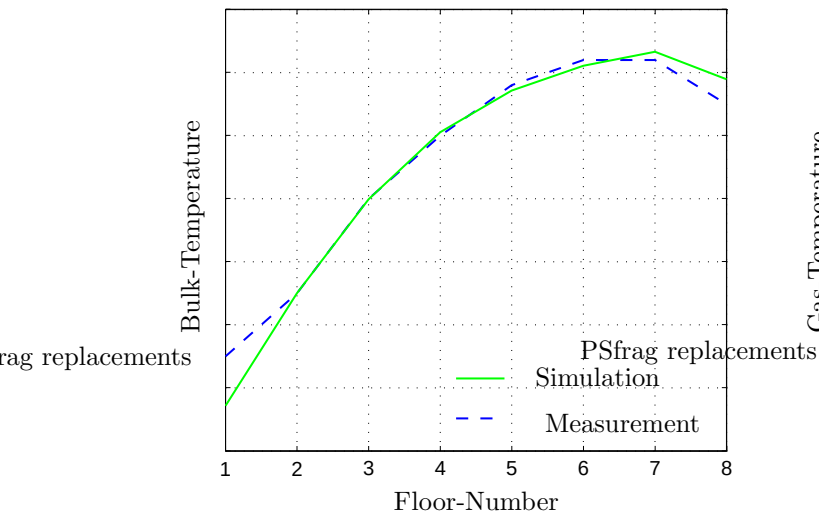


Figure 3: Stationary temperature-profile for the bulk

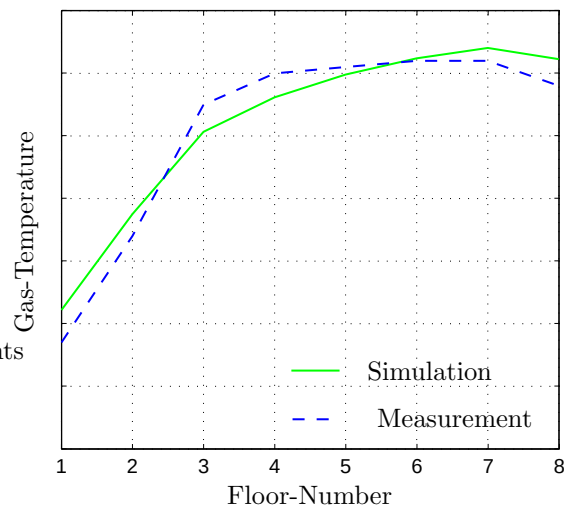


Figure 4: Stationary temperature-profile for the gas

For validation of the transient behaviour a step-increase of the rotation speed of the rotating-shaft was carried out. In this case the temperature decreases, as can be seen in Figure 5 for the bulk-temperature in the 5th floor and in Figure 6 for the gas-temperature in the 5th floor.

For a step-wise reduction of the oil-massflow it is plausible to obtain a decreasing temperature, which can also be seen in Figure 7 for the bulk-temperature of the 5th floor and in Figure 8 for the gas-temperature. The Comparison of measurement- and simulation-data shows very good accuracy of the thermodynamic model, although the measurement seems heavily disturbed.

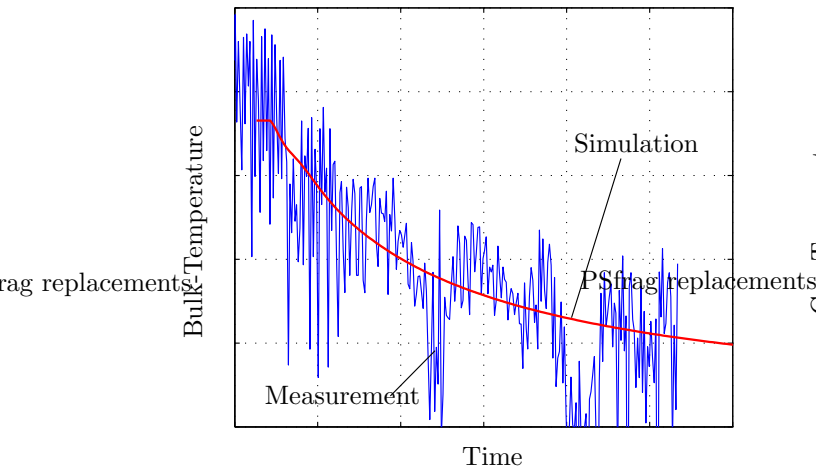


Figure 5: Bulk temperature of floor 5 over time for a stepwise change of the shaft-rotation-speed

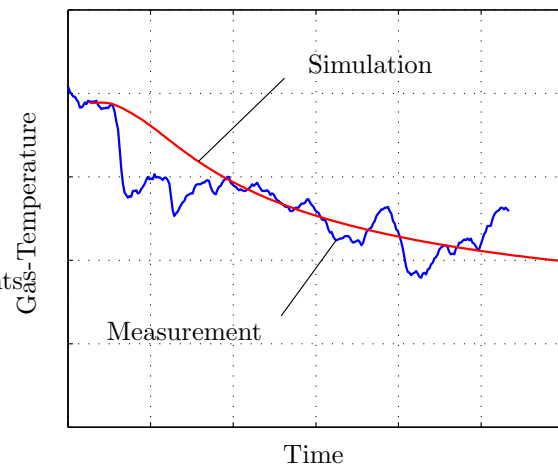


Figure 6: Gas temperature of floor 5 over time for a stepwise change of the shaft-rotation-speed

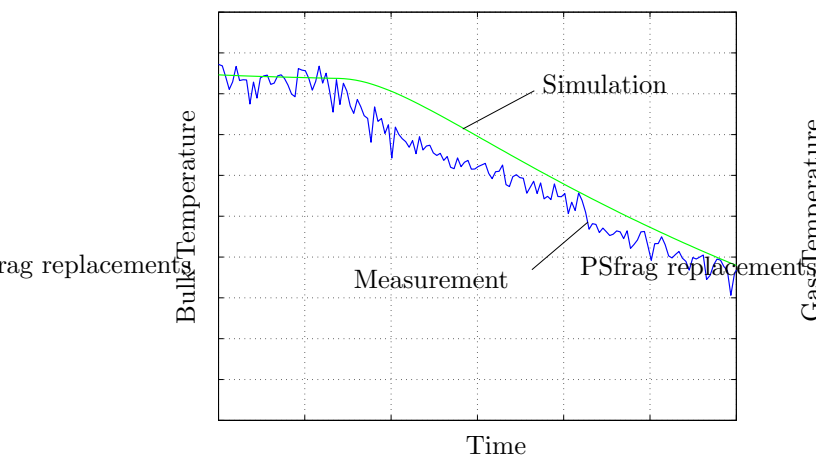


Figure 7: Bulk-temperature of floor 5 over time for a stepwise change of the oil mass-flow

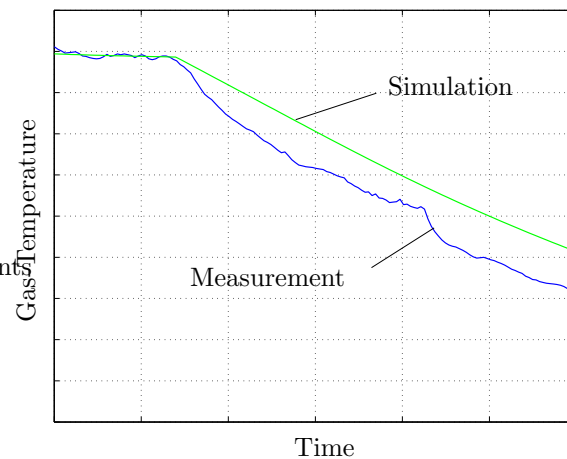


Figure 8: Gas-temperature of floor 5 over time for a stepwise change of the oil mass-flow

Validation of the Chemical Model:

Validation of the chemical model is a tedious thing as continuous measurement for bulk-concentrations is usually not available. Therefore, several samples of the bulk-media have to be taken by hand during the transient change of the concentration. The experiment showed that one sample at a time is not enough for a representative measurement-series. This can be explained by the high degree of inhomogeneity of the bulk-media and has to be encountered by multiple sample-taking per measurement-point. These data were not yet accessible and therefore measurement-data-based validation of the dynamic part of the chemical model was not possible yet. Nevertheless, the dynamic behaviour does not look fully plausible to process-experts and from that point of view the validation failed.

5. Conclusions

Modelling a roasting-process in a multiple-hearth-furnace in the way described above can be used advantageously. It results in an easy to handle mathematical model that can be implemented in simulation-software without problems. The comparison of measurement- and simulation-data showed excellent consistency of the transportation and thermodynamical part of the model. Validation of the chemical part of the model failed, therefore further work is in progress to replace the simple rate-law by a better and more enfolding rate-law.

6. Nomenclature

A	m^2	floor area	M_i	g/mol	molecular mass of comp. i
α_{GS}	W/m^2K	heat transference coeff. g/s	$\dot{m}_{eff}$	kg/s	effective mass-flow
α_{DS}	W/m^2K	heat transference coeff. ds	$\dot{m}_s$	kg/s	actual mass-flow
b_W, b, h_D	m	geometric parameters	n_{ds}	$1/min$	rotation-speed of ds
c_{ed}	mol/m^3	educt concentration	R	$\frac{J}{mol K}$	universal gas constant
c_{prod}	mol/m^3	product concentration	r_i	m	inner radius
c_i	mol/m^3	concentration of comp. i	r_o	m	outer radius
c_p	$\frac{J}{mol K}$	specific heat capacity	r_V	$\frac{mol}{s m^3}$	volume-specific reaction-rate
E	J/mol	activation energy	ρ_s	kg/m^3	density of the bulk-media
f_{reflux}	1	reflux-factor	t	s	time
f_{trans}	1	transportation-factor	T_{DS}	K	temperature of drive-shaft
h_g	m	height of gas-layer	T_g	K	temperature of gas
h_s	m	height of bulk-layer	T_s	K	temperature of bulk
$h_{M,i}$	J/g	melting heat for comp. i	T_E	K	temperature of environment
ΔH_{ev}	J/kg	heat for evaporation	u_s	m/s	velocity of incoming bulk
$(-\Delta H_r)$	J/mol	heat of reaction	u_g	m/s	velocity of incoming gas
k	$1/s$	drying constant	v_I	$\frac{kg}{s kg}$	drying-velocity in period I
k_0	$1/s$	kinetic const. fwd reaction	V_s	m^3	bulk-volume
k_0'	$\frac{m^3}{s \cdot mol}$	kinetic const. bwd reaction	w_{H_2O}	kg/kg	weight-fraction of H_2O
λ_W	$\frac{W}{m K}$	heat conductivity	$w_{I,II}$	kg/kg	boundary for periods I/II

7. References

1. Krischer, O. and Kast, W., Die wissenschaftlichen Grundlagen der Trocknungstechnik, Springer Berlin [u.a.], 1992
2. Mueller-Erlwein, E., Chemische Reaktionstechnik, B.G.Teubner Stuttgart-Leipzig, 1998.
3. Chen, P., Pei, C.T.D., A mathematical model of drying processes, In: Int. J. Heat Mass Transf. 32 (1989), 297 – 310.
4. Shene, C., Cubillos, F., Perez, R. Alvarez, P.I., Modelling and Simulation of a Direct Contact Rotary Dryer, In: Drying Technology 14 (1996), 2419-2433.

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